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# PALEOMAGNETISM OF THE TYPE PLIOCENE/PLEISTOCENE BOUNDARY SECTION AT SANTA MARIA DE CATANZARO, ITALY, AND THE PROBLEM OF POST-DEPOSITIONAL PRECIPITATION OF MAGNETIC MINERALS

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In order to assist in global correlation of the Pliocene/Pleistocene boundary, a paleomagnetic survey has been made of the world type section for the boundary, in Calabria, Italy. A total of 116 specimens from 36 sites were collected from the section, which as sampled is about 30 m thick. All sites possess stable remanent magnetism and northern hemisphere virtual geomagnetic poles, consistent with deposition during a period of normal polarity. If this period was during the Gilsa event, a minimum sedimentation rate of 17 cm per 1000 years would be required. Since a normal polarity chemical overprint, acquired during the last 0.69 m.y. could also explain the data, thermomagnetic analyses were made of selected samples. The results show the presence of highly unstable superparamagnetic material, which is most likely a product of post-depositional chemical precipitation. This observation enhances the possibility that any longer-duration precipitation could have created coarse and thus magnetically more stable components, the effects of which could not be readily distinguished from the original remanent magnetism.

The difficulties of distinguishing between original depositional remanent magnetism and post-depositional chemical remanent magnetism, in outcropping marine sediments, as well as the large range of possible natural causes of the latter, is summarized in the form of a discussion of feasible Eh and pH changes occurring between original deposition at upper bathyal dephts and final sampling above sea level. It is considered that, in the absence of experimental means to distinguish the roles of original and chemically overprinted paleomagnetic signals, similarity of magnetic polarity stratigraphy between sections of sediments representing different paleoenvironments and sedimentation rates is a necessary if insufficient requirement for diagnosis of real geomagnetic behavior during deposition, as opposed to post-depositional effects.

## 1. Introduction

Past reversals of the geomagnetic field are commonly reflected in the paleomagnetic properties of igneous and sedimentary rock sequences. K–Ar dating and paleomagnetic studies of igneous rocks have led to establishment of the geomagnetic polarity time scale for the past 5 m.y. [1]. The existence of this time scale has facilitated, by recognition of corresponding marine magnetic anomaly patterns, successful dating of oceanic crust [2] produced during sea-floor spreading [3]. Similarly, paleomagnetic dating of Pleistocene through Pliocene deep-sea sedimentary cores is now an established method [4] employing recognition of the known polarity sequence; but because of ambiguities, it is ideally supplemented by microfaunal data [4,5]. These successes have led to application of the same principles to dating of outcropping Late Cenozoic marine and non-marine sediments [6–12], and substantial efforts are being made, using sediments, to establish a polarity stratigraphy for the Mesozoic era [15–18]. In the former category, while several interpretations are unambiguous in terms of that segment of the polarity scale assigned to the data, others are unsatisfactory. For example, New Zealand sediments thought to have been deposited during the Gilbert reversed polarity epoch [13] and Italian sediments [4] confined to the Matuyama reversed polarity epoch are all mostly normally magnetized. Interpretation of results from Czechoslovakian sediments [8] requires a polarity reversal history for the past 0.5 m.y. quite incompatible with independent information. Despite an

attempt to develop innovative means to recognize the polarity reversals of the past 5 m.y. in sediment cores from leg 13 of the Deep Sea Drilling Project [19,20], the results nevertheless show that cores spanning most of the Pleistocene and Pliocene are normal polarity throughout [11]. Until now, only Kent [12]has shown that particular Late Cenozoic sedimentary sequences do not provide meaningful polarity stratigraphy: the remanent magnetism of the Japanese sediments measured was completely dominated by viscous components.

It is the purpose of this paper to present the results of a paleomagnetic study of the type section for the Pliocene/Pleistocene boundary at Santa Maria de Catanzaro, Calabria, Italy. Dating of this section has much significance for Late Cenozoic global stratigraphy. We shall, on the basis of our analyses, draw attention to serious difficulties in the interpretation of paleomagnetic dating of young marine sedimentary sequences.

## 2. Geology of the section and previous work

The lower part of the Santa Maria de Catanzaro section (Fig. 1), which is Late Pliocene in age, consists of blue-grey mudstones and siltstones. These grade upward into a coarse sandy layer marking the base of the Calabrian stage, which is the base of the type Pleistocene. The sand termed Horizon G by Gignoux [21] is succeeded by a sequence of siltstones and sandstones. The section is capped by silty mudstones. Selli [22, 23] proposes that the basal Pleistocene sands are turbidites, which Lamb [24], however, believes were emplaced in bathyal depts. In contrast, Bandy and Wilcoxon [25] interpret the benthic foraminiferal assemblage to be consistent with deposition at upper bathyal depths (between 200 and 500 m depth). Most recently, Drooger [26], using the benthonic foraminifera, and Sissingh [27], using ostracoda propose depositional depths of only 40–150 m. Similarly, the earlier turbidite idea for the lower part of the Calabrian sequence has, as the result of very detailed sedimentary studies by Brolsma and Meulenkamp [28], been replaced by a model involving simple deposition in shallow high-energy environment much like the present Straits of Messina [29]. The underlying Late Pliocene sediments and overlying finer-grained Calabrian sediments were probably deposited in slightly deeper waters. Thus, a possibly wide range of paleoenvironments are represented by the sediments in the section.

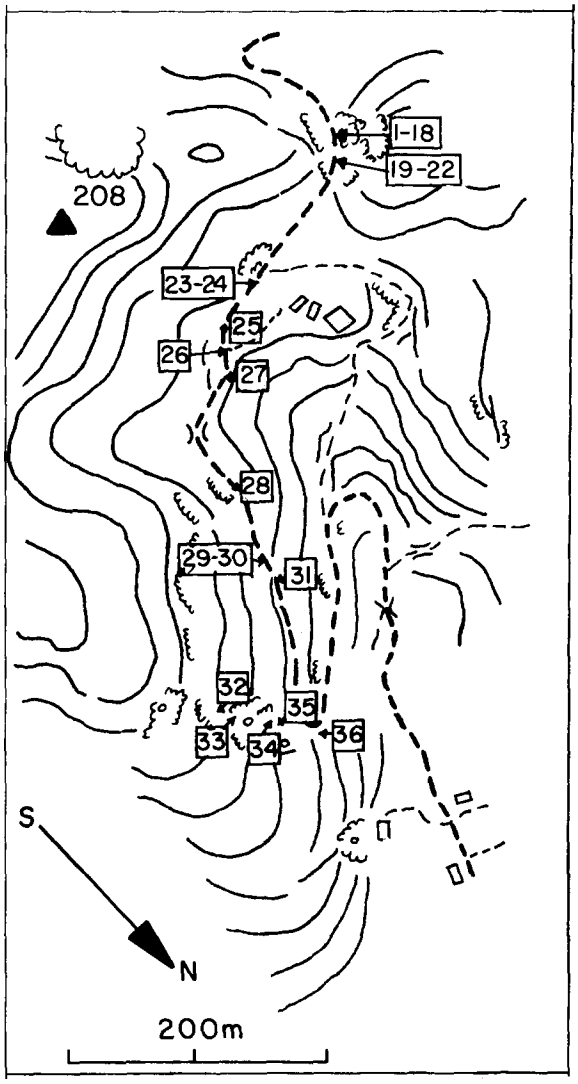



Fig. 1. Map showing location of sample sites (1 to 36) at the Santa Maria de Catanzaro section. Base map from Nakagawa et al. [14]. Detailed location given by many authors [14,21–29]. The coordinates of the bottom left-hand corner of the map are: latitude = 38°54′N, longitude = 16°38′E.

An approximate age of 1.8 m.y. has been proposed for the Pliocene/Pleistocene boundary, on the basis of the correlation of the *Globorotalia truncatulinoides* datum, slightly above the base of the Pleistocene at Santa Maria de Catanzaro [29], with the base of the Gilsa polarity event [30,31] in tropical deep-sea sedi-

mentary cores. The Pliocene/Pleistocene boundary therefore occurs during the Matuyama reversed epoch within or close to the Gilsa normal polarity event, which extends from 1.61 to 1.79 m.y. ago [1]. Nakagawa et al. [6,14] have presented paleomagnetic polarities (following alternating magnetic field cleaning of specimens), for a total of 18 sites in the section. Twelve of the sites are presented as normal polarity, and the favored interpretation is that Middle Matuyama sediments exist in the section, consistent with the 1.8 m.y. age estimate [30,31].

## 3. Methods, results, and discussion

We have collected three separately oriented cores

TABLE 1

Paleomagnetic results Santa Maria de Catanzaro

| No. | Strat. level | Lithology | $N$ | $D^\circ$ | $I^\circ$ | $J$ | $R$ | $\phi'$ | $\theta'$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 30.6 | muddy siltstone | 2 | 12.8 | 59.8 | 2.21 | 1.9711 | 86.7 | 79.7 |
| 2 | 30.3 | silty mudstone | 3 | 40.4 | 64.4 | 0.15 | 2.7303 | 77.7 | 59.3 |
| 3 | 30.0 | silty mudstone | 3 | 14.9 | 53.9 | 6.31 | 2.9949 | 118.0 | 77.5 |
| 4 | 29.7 | silty mudstone | 3 | 14.7 | 56.8 | 20.80 | 2.9922 | 104.3 | 78.3 |
| 5 | 29.1 | silty mudstone | 3 | 19.8 | 54.1 | 0.18 | 2.9950 | 111.7 | 73.7 |
| 6 | 28.7 | silty mudstone | 3 | 17.6 | 51.2 | 0.65 | 2.9909 | 123.8 | 74.4 |
| 7 | 28.3 | mudstone | 3 | 8.5 | 46.3 | 0.54 | 2.9593 | 159.1 | 77.4 |
| 8 | 28.0 | silty mudstone | 3 | 17.3 | 49.4 | 0.68 | 2.9873 | 129.3 | 73.7 |
| 9 | 27.7 | silty mudstone | 3 | 6.0 | 36.3 | 0.24 | 2.6379 | 178.1 | 71.4 |
| 10 | 27.3 | silty mudstone | 3 | 18.3 | 54.7 | 0.07 | 2.9950 | 111.1 | 75.1 |
| 11 | 26.9 | silty mudstone | 3 | 33.5 | 66.9 | 0.04 | 2.9727 | 69.4 | 63.5 |
| 12 | 26.6 | silty mudstone | 3 | 21.9 | 56.9 | 0.89 | 2.9961 | 101.1 | 72.7 |
| 13 | 26.3 | silty mudstone | 3 | 9.8 | 64.6 | 0.04 | 2.9832 | 53.3 | 78.9 |
| 14 | 26.0 | silty mudstone | 3 | 11.0 | 59.1 | 0.09 | 2.9940 | 90.3 | 81.2 |
| 15 | 25.6 | silty mudstone | 3 | 37.9 | 36.5 | 0.26 | 2.8056 | 123.5 | 52.8 |
| 16 | 25.3 | muddy siltstone | 3 | 17.3 | 39.9 | 1.10 | 2.8081 | 146.9 | 68.7 |
| 17 | 24.9 | muddy siltstone | 3 | 13.6 | 56.5 | 0.67 | 2.9963 | 106.6 | 79.2 |
| 18 | 24.5 | muddy siltstone | 3 | 6.2 | 53.0 | 0.83 | 2.9901 | 145.3 | 83.3 |
| 19 | 24.0 | muddy siltstone | 3 | 8.1 | 39.0 | 1.15 | 2.9775 | 170.2 | 72.6 |
| 20 | 23.5 | silty mudstone | 3 | 8.7 | 50.3 | 1.49 | 2.9945 | 147.6 | 80.1 |
| 21 | 22.9 | silty mudstone | 3 | 15.7 | 53.6 | 0.07 | 2.9875 | 118.1 | 76.7 |
| 22 | 22.3 | silty mudstone | 3 | 59.9 | 70.8 | 0.07 | 2.1474 | 62.5 | 47.0 |
| 23 | 12.7 | sandy siltstone | 3 | 287.5 | 58.7 | 0.02 | 2.9060 | 311.9 | 35.1 |
| 24 | 11.9 | sandy siltstone | 3 | 5.4 | 57.6 | 0.01 | 2.9579 | 101.7 | 85.7 |
| 25 | 11.5 | muddy siltstone | 3 | 0.5 | 41.4 | 0.01 | 2.9055 | 194.1 | 75.8 |
| 26 | 9.5 | sandy siltstone | 3 | 308.2 | 48.1 | 0.01 | 2.7013 | 290.2 | 46.5 |
| 27 | 8.9 | sandy siltstone | 3 | 331.9 | 32.8 | 0.01 | 2.7915 | 254.5 | 58.3 |
| 28 | 6.9 | sandy siltstone | 3 | 351.3 | 49.9 | 0.01 | 2.7502 | 243.1 | 79.7 |
| 29 | 5.4 | muddy siltstone | 3 | 349.6 | 32.9 | 0.02 | 2.8322 | 223.2 | 68.0 |
| 30 | 4.9 | muddy siltstone | 3 | 342.5 | 57.2 | 0.02 | 2.9744 | 290.7 | 76.2 |
| 31 | 4.3 | muddy siltstone | 3 | 2.0 | 58.8 | 0.01 | 2.9194 | 59.8 | 87.8 |
| 32 | 3.3 | mudstone | 3 | 55.7 | 50.6 | 0.01 | 2.5666 | 96.9 | 44.4 |
| 33 | 2.7 | mudstone | 3 | 293.3 | 57.3 | 0.01 | 2.4822 | 308.0 | 38.6 |
| 34 | 2.5 | mudstone | 3 | 6.9 | 66.1 | 0.01 | 2.9046 | 39.2 | 78.4 |
| 35 | 2.0 | mudstone | 2 | 337.7 | 67.6 | 0.01 | 1.9351 | 331.5 | 69.9 |
| 36 | 0.0 | silty sandstone | 3 | 26.2 | 45.1 | 0.01 | 2.9657 | 125.6 | 65.3 |

No. = site number (see Fig. 1 for location); Strat. level = level of sample site in meters above lowest sample site (Fig. 1); $N$ = number of separately oriented samples; $D^\circ$ and $I^\circ$ = mean declination and inclination of remanent magnetism at sites ($D$ is degrees east; $I$ = degrees below horizontal); $J$ = mean intensity of magnetization in emu $g^{-1} \times 10^6$; $R$ = resultant vector, assigning unit vector per sample; $\phi'$ and $\theta'$ = longitude and latitude of virtual geomagnetic pole, in degrees east of Greenwich and in degrees north, respectively. $D$, $I$, $J$, $R$, $\phi'$ and $\theta'$ are results following magnetic cleaning of all specimens in an alternating field of 50 oersteds.

from each of a total of 36 separate sites, all of which were sampled after removing up to 1 m of weathered surface. The locations of the sites are given in Fig. 1, and we include a lithological description of the sites in Table 1. The natural remanent magnetism (NRM) of all specimens is normal polarity. Treatment in alternating magnetic fields shows that the NRM is stable, with median destructive fields ranging from 75 to as high as 400 oersteds, and averaging 200 oersteds. Demagnetization in fields as high 1000 oersteds does not yield a reversed polarity. A small unstable component is minimized by alternating magnetic fields of only 50 oersteds. Results for all sites following such demagnetizing treatment are given in Table 1. Conventional interpretation of these data would suggest that the entire section was deposited during a period of normal geomagnetic polarity, conceivably during the Gilsa event, necessitating a minimum sedimentation rate of about 17 cm per 1000 years.

Progressive thermal demagnetizing of twenty selected specimens, spaced regularly throughout the section was carried out in a 40-gamma maximum field at temperatures up to 600°C, in a nitrogen atmosphere. At temperatures above 350°C, an order of magnitude increase in the resulting intensity of magnetization ($J$) of all twenty specimens suggested production of new magnetic phases, by the heating process. This increase in $J$ was not accompanied by significant changes in the scatter of directions of remanent magnetism. Below 350°C, a completely unstable magnetic component was produced, with a relaxation time so short that the polarity of any specimen reversed as rapidly as the specimen could be inverted by hand. It was verified that the unstable component was not produced in the oven, but by the ambient magnetic field in the laboratory, between the oven and spinner magnetometer. This component is not present in the NRM of specimens or in specimens treated in alternating magnetic fields. Saturation magnetization of specimens thermally demagnetized at 100°C to 300°C (Fig. 2) shows an increasing production of the unstable magnetic phase by the heating process. We interpret this to be the result of increasing dehydration of an iron oxhydroxide (probably goethite) to produce a superparamagnetic component. Exactly the same behavior has been observed in specimens from twelve successive sites at La Castella (80 km to the east), which is thought to be the stratigraphic equivalent of the Santa Maria de Catanzaro section.

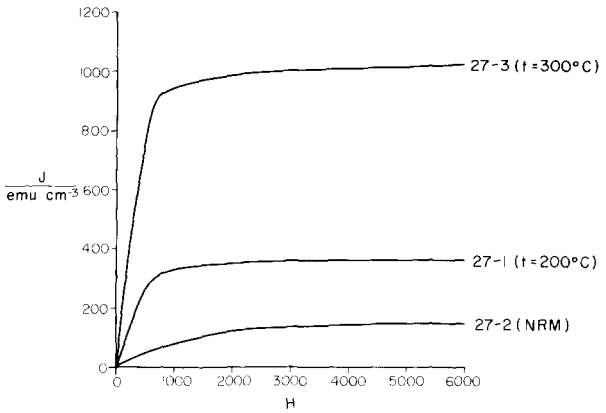



Fig. 2. Saturation magnetization curves of three specimens from Santa Maria de Catanzaro section: $J$ = intensity of magnetization in emu $cm^{-3}$; $H$ = ambient geomagnetic field in gauss. The three curves are for different specimens from the same site subjected to no heating, heating to 200°C, and heating to 300°C, respectively. The curves are interpreted to show an increasing production of a magnetic phase with increasing temperature causing a corresponding increasing dehydration.

Although it has long been realized that post-depositional precipitation of magnetic minerals can be an important factor in the paleomagnetism of certain sediments [32,33], insufficient consideration has been given to the relationship between environments which a sediment can be expected to encounter (between its original deposition below sea level and eventual sampling) and the corresponding range of precipitated minerals. An Eh–pH diagram provides a convenient summary of the relationship between thermodynamic stability of iron solid phases and acid-base redox reactions [34,35]. Fig. 3A shows the Eh–pH characteristics of several environments in which the Eh is controlled by either the presence of 0.01 to 6.0 ml/l dissolved oxygen or of $10^{-6}$ to $10^{-3}$ molar hydrogen sulfide: these conditions are typical of a wide range of marine environments, as indicated. In Fig. 3B and C we show the same series of environments superimposed on the stability field of hydrous iron oxides, iron oxides, and iron sulfide. A possible relevant sequence of events for the Santa Maria de Catanzaro deposits could include the following: particle deposition in outer shelf to upper bathyal depths; perhaps oscillations between oxic and anoxic conditions; and initial elevation to very shallow depths, with possible lagoonal and even fringing reef conditions. After elevation above sea level, erosion and exposure to water table fluc-

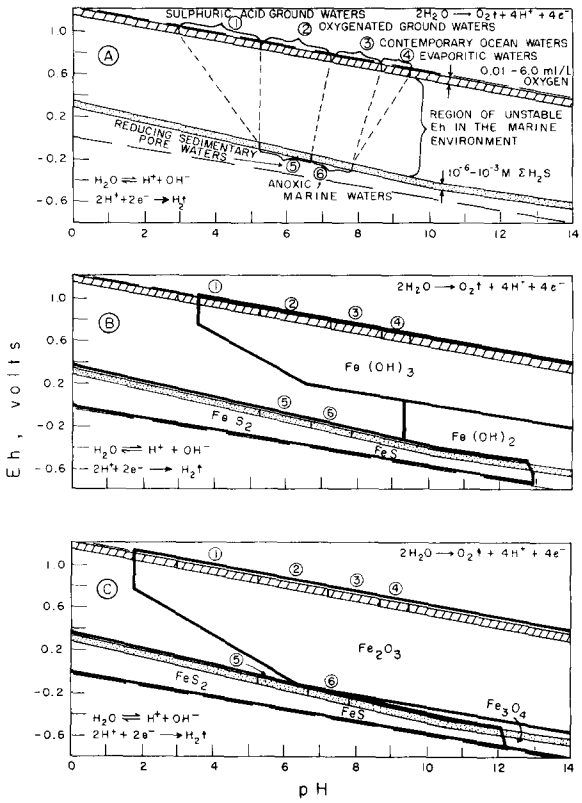



Fig. 3. A. Eh–pH diagram of natural aqueous systems bounded by the stability limits of water (dashed lines) and showing the Eh control bands of 0.01 to 6.0 ml/l dissolved oxygen and of $10^{-6}$ to $10^{-3}$ molar hydrogen sulfide. B. Eh–pH domains for the metastable hydrous oxides of iron and for ferrous sulfides. C. Eh–pH domains for the stable oxides of iron and for ferrous sulfides. In the region of the Eh–pH diagram to the left of the solid phase boundaries in B and C the solubility of iron exceeds $10^{-6}$ molar. In all three diagrams the six types of environments are identified in terms of their Eh–pH fields. These are labeled by name in diagram A only.

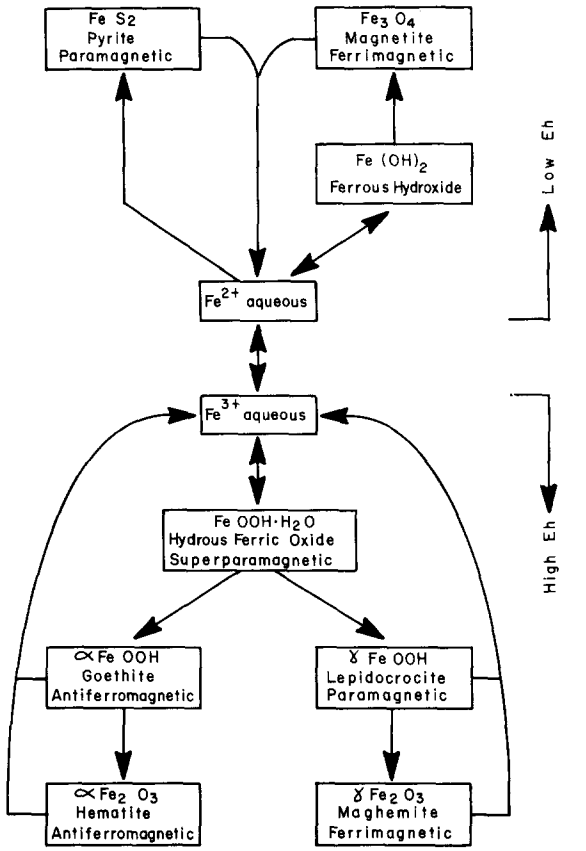



Fig. 4. Potential paths for the chemical alteration of a magnetic mineral ($Fe_3O_4$) following its original deposition, during an excursion between oxic and anoxic portions of the environmental Eh–pH regime. The arrows are used to indicate the reversability of some of the processes.

tuations including periods when evaporation could be dominant. Additional variations in the Eh–pH conditions of those sediments are likely to have occurred due to the effect of vegetation cycles on the groundwater chemistry. Thus the range of possible magnetic mineral precipitates could be large (Fig. 3).

Fig. 4 illustrates some of the chemical cycling of iron which may occur during cyclic excursions between the extreme oxygenated and sulfide conditions (Fig. 3). For example, consider an initially deposited detrital magnetite particle which may be converted to $FeS_2$ and/or FeS upon exposure to reducing conditions (regions 5 and 6 in Fig. 3). Subsequent oxygenation of this sediment will transfer the iron through the aqueous phase and into the series of hydrous oxides with the ultimate production of the oxides (regions 1–4 in Fig. 3). The magnetic properties resulting from this process will depend on the solid phases which are formed, and on the Earth's magnetic field prevailing at the time of their formation. Samples which are several hundred thousand years old can clearly be subjected to repeated Eh–pH excursions. The chemical alteration of the solid phases will destroy the original detrital remanent magnetization and yield a magnetic overprint indicative of the final phase of crystallization. If this process continued during the last 0.69 m.y. then a normal polarity would be ex-

pected, particularly if the original detrital remanent magnetism is of low intensity. Consideration of the relative stability of solid phases on an Eh–pH diagram is useful for specifying the equilibrium condition, but the specific path or rate of the process leading to equilibrium can be complex [36,37]. Some of these processes, such as the oxidation of iron sulfides can be readily observed and characterized kinetically [38].

Nagata [33] has presented a model for the acquisition of chemical remanent magnetism in sediments, in which three stages are probable in the conversion of an amorphous hydrous oxide to a stable crystalline oxide. Stage I consists of very small particles which are less than the critical magnetic domain size, about 20 Å [39], and are therefore superparamagnetic. Stage II is of a particle size comparable to the domain size and thus results in magnetically stable single-domain particles. Stage III represents large multi-domain particles which display paramagnetism. The thermally induced behavior we observed can be interpreted as being due to the presence of hydrous ferric or ferrous phases remaining from the chemical processes suggested in Fig. 4. Upon partial dehydration, these phases produced stage I (superparamagnetic particles) which progressed to stage II with further heating.

## 4. Conclusions

The major aspect of our observations relevant to paleomagnetic dating, or polarity stratigraphy of outcropping marine sediments, is the lack of a magnetically stable end member. There is no reason to believe that the dominance of any particular phase reveals the required original ambient geomagnetic field polarity. In the case of the data presented in this paper (Table 1), we can be confident that we have diagnosed the presence of a fine precipitated unstable phase which was not observable in the NRM of the specimens, but which became dominant upon dehydration. Had natural crystallization taken place for a longer period, however, to produce particles larger than the critical superparamagnetic values, then in all probability conventional paleomagnetic methods could only have provided strong evidence of a stable component. Given the possibility that such solid phase re-cycling could occur, and recur, at any time following deposition (Fig. 4), then plausible environmental models could be invoked to yield magnetic polarities and virtual geomagnetic poles quite unrelated to the original parameters. The fact that the cores from leg 13 of the Deep Sea Drilling Project are normal polarity throughout [19,20] despite an age range of 4–5 m.y., may be due to their location in the Mediterranean, which has experienced substantial Eh and pH changes during the past 6 m.y. [40]. Kobayashi and Nomura [41] have shown, for example, that sulphides have been precipitated in the bottom sediments of the Sea of Japan, consistent with the anoxic environment present. Such geochemical extremes may not have been common in the major ocean basins during the Pleistocene at least, judging by the common agreement between the established polarity time scale and much published paleomagnetic data for deep-sea sedimentary cores. No similar inference can as yet be made for older marine sediments: the relative contributions of detrital materials from the continents to the marine realm has almost certainly increased since the Miocene in many ocean regions, and this would tend to minimize the relative importance of chemical precipitates. The problem of chemical overprinting is far more critical in polarity dating of Late Cenozoic sediments, than it is in the determination of Upper Paleozoic pole positions [42,43], because a lag between deposition and final polarity acquisition of the order of $10^6$ to even $10^7$ years is of minimal geological significance for the Upper Paleozoic. Polarity stratigraphy in older marine sequences could be adversely affected by the Eh and pH variability (Fig. 3), which could be diachronous in their effects. At the same time, however, it is quite possible that original detrital remanent magnetism can be preserved in sediments which have undergone minimal environmental changes, as reported by Reeve and Helsley [44], Roy and Park [45], and others, for Mesozoic sediments.

It remains to be seen whether or not experimental means can be devised to remove the effect of post-depositional precipitation of magnetic minerals in Late Cenozoic sediments, or to prove conclusively that such precipitation has not taken place to any significant extent. It appears that at present a major criterion for some acceptance of paleomagnetic dating of such young sedimentary sequences will be whether or not the inferred polarity stratigraphy agrees with results of independent dating methods, such as fission track or K–Ar dating of suitable contempora-

neously deposited materials, including volcanic ash. In addition to the problems which we have discussed here, there also exists the possibility of magnetic signature distortion by simple dynamic processes during deposition [46], and we therefore propose that fine structure of the geomagnetic polarity history cannot as yet be reliably derived from marine sediments, in particular, but also from sediments deposited in lakes with variable chemical histories, unless relevant data is derived from several well-separated sections of different lithologies, and all are independently dated [47]. The recent observations of Larson et al. [48], which show that polarities in Late Tertiary sediments from Baja California depend on their color, even though original deposition was contemporaneous, provides strong emphasis for this point.

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